

GRANDE PRAIRIE, AB, Canada

WMO: 719400

Lat: 55.180N

Lon: 118.880W

Elev: 669

StdP: 93.54

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.9	-30.7	-39.5	0.1	-34.9	-35.1	0.1	-30.5	13.4	2.7	12.7	2.5	1.2	330	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	27.4	16.5	25.6	15.8	24.0	15.1	18.0	25.1	16.9	23.6	15.9	22.2	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.9	20.6	14.3	11.0	19.3	13.2	10.3	18.3	53.6	25.1	50.0	23.5	46.9	22.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.7	10.0	8.5	DB	-39.2	30.8	4.3	1.8	-42.3	32.1	-44.9	33.2	-47.3	34.2	-50.5	35.6
(5)				WB	-39.3	19.9	4.3	1.2	-42.4	20.8	-44.9	21.5	-47.3	22.2	-50.4	23.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-13.1	-10.8	-5.8	3.5	10.2	14.3	16.4	15.2	10.5	3.6	-6.1	-12.0
(7)		DBStd	12.46	9.88	8.31	7.86	5.08	4.30	2.89	2.79	3.23	4.03	4.70	7.39	8.71
(8)		HDD10.0	3454	717	584	490	199	49	3	1	2	42	201	484	683
(9)		HDD18.3	5904	975	817	748	446	252	124	71	104	235	456	733	941
(10)		CDD10.0	617	0	0	0	3	56	132	201	164	57	4	0	0
(11)		CDD18.3	26	0	0	0	0	1	3	13	9	0	0	0	0
(12)		CDH23.3	496	0	0	0	1	30	73	199	157	34	1	0	0
(13)		CDH26.7	88	0	0	0	0	3	9	39	32	4	0	0	0
(14)	Wind	WSAvg	3.3	2.7	2.7	3.2	3.9	4.1	3.9	3.7	3.3	3.5	3.4	2.8	2.7
(15)	Precipitation	PrecAvg	477	36	23	21	21	39	79	70	65	44	23	29	29
(16)		PrecMax	720	120	68	60	48	95	176	176	186	129	58	59	98
(17)		PrecMin	252	7	3	3	2	3	6	9	12	4	6	1	1
(18)		PrecStd	110	25	16	13	13	25	49	44	42	30	12	17	19
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.0	10.1	20.9	26.7	27.8	30.4	29.7	27.0	20.3	11.6	7.7
(20)			MCWB	2.8	2.3	5.1	10.3	13.7	16.0	18.3	17.7	15.6	11.6	6.2	4.0
(21)		2%	DB	4.7	4.2	7.4	17.0	23.4	25.5	27.4	27.1	23.8	15.8	7.1	5.1
(22)			MCWB	2.1	1.4	3.1	8.3	12.5	15.0	17.0	17.2	14.4	9.2	3.8	2.1
(23)		5%	DB	3.1	2.7	5.6	14.0	21.2	23.5	25.5	24.9	21.1	13.1	4.8	2.8
(24)			MCWB	0.8	0.3	2.0	6.6	11.3	14.1	16.4	15.9	13.3	7.4	2.0	0.4
(25)		10%	DB	0.7	0.7	4.0	11.6	18.9	21.5	23.7	22.9	18.4	11.0	3.1	0.4
(26)			MCWB	-1.0	-1.2	1.0	5.3	10.3	13.2	15.6	15.1	11.8	6.2	0.9	-1.5
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.7	5.3	10.9	14.8	17.4	19.7	19.5	16.6	12.4	6.2	4.0
(28)			MCD B	5.9	5.4	9.1	19.8	24.2	24.8	26.6	26.3	24.7	19.4	10.4	7.3
(29)		2%	WB	2.2	1.5	3.6	8.7	13.2	16.1	18.4	17.9	15.0	9.6	4.2	2.2
(30)			MCD B	4.5	4.1	6.9	16.3	21.6	23.3	25.1	25.1	22.5	14.7	7.1	4.8
(31)		5%	WB	0.9	0.4	2.2	7.0	11.9	15.0	17.2	16.8	13.6	8.1	2.3	0.5
(32)			MCD B	2.9	2.6	5.2	13.1	19.7	21.7	23.7	23.3	20.1	12.2	4.3	2.7
(33)		10%	WB	-0.9	-1.2	1.1	5.7	10.8	14.0	16.1	15.7	12.2	6.6	1.0	-1.5
(34)			MCD B	0.8	0.7	3.9	10.9	17.9	20.1	22.2	21.6	17.6	10.3	2.8	0.3
(35)	Mean Daily Temperature Range	MDBR	9.6	10.4	10.3	10.9	12.4	11.5	11.7	12.3	12.2	10.2	8.7	9.3	
(36)		5% DB	MCD BR	10.5	11.2	11.0	16.0	16.6	15.2	15.6	16.3	17.7	14.2	10.3	11.1
(37)			MCWBR	8.5	9.0	7.9	9.1	8.2	7.2	7.3	7.7	9.3	8.7	7.7	9.2
(38)		5% WB	MCD BR	10.4	10.8	10.7	14.6	15.0	13.4	14.1	14.8	16.7	13.7	9.9	11.1
(39)			MCWBR	8.6	8.9	7.9	8.7	7.7	6.7	7.1	7.4	9.0	8.6	7.7	9.3
(40)	Clear-Sky Solar Irradiance	taub	0.238	0.253	0.304	0.327	0.347	0.350	0.350	0.350	0.311	0.286	0.247	0.236	
(41)		taud	2.366	2.413	2.326	2.390	2.393	2.401	2.432	2.418	2.534	2.564	2.451	2.271	
(42)		Ebn at Noon	739	859	872	899	896	895	888	865	863	810	725	636	
(43)		Edh at Noon	49	68	95	103	109	110	104	99	77	59	45	42	
(44)	All-Sky Solar Radiation	RadAvg	0.75	1.67	3.10	4.62	5.46	5.97	5.87	4.79	3.31	1.80	0.82	0.54	
(45)		RadStd	0.07	0.13	0.24	0.28	0.39	0.49	0.40	0.40	0.31	0.19	0.07	0.04	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page